An Effective Adaptive Filter to Reduce Motion Artifacts from ECG Signals Using Accelerometer

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ABSTRACT
Electrocardiogram (ECG) is most commonly used for the diagnosis of heart disease. Physicians utilize high quality ECG to interpret and identify physiological and pathological phenomena. However, motion artifacts are caused by the electrode-skin impedance with electrode motions that are the results of a subject's movement, so it is a classical problem to reduce motion artifacts from ECG signals during real-time heart rate measurements. To the best of our knowledge, Block Least Mean Square (BLMS) algorithm has not been considered in the context of the use of accelerometer measurement data in ECG signals as a reference signal on reducing motion artifacts. Therefore, in this paper, we propose a 3-axis accelerometer to measure the acceleration signal of the movement of the trunk as the reference input of the adaptive filter and the optimal weight of the adaptive filter is adjusted by the BLMS algorithm. Finally, we have applied this algorithm on ECG signals from the subject and compared its performance with the conventional Least Mean Square (LMS) algorithm. The results show that the performance of the BLMS algorithm is superior than the LMS algorithm. And, we have found the R wave of the filtered ECG was clearly appeared.

CCS Concepts
Applied computing → Life and medical sciences

Keywords
ECG; Motion artifacts; Accelerometer; Adaptive filter; BLMS

1. INTRODUCTION
Electrocardiogram (ECG) is widely used to diagnose heart disease, such as arrhythmia, coronary heart disease, high blood pressure and so on. However, ECG signals are easily corrupted by various noises. Common noises mainly have power line interference, motion artifacts, myoelectric interference, and baseline drift with respiration. Motion artifacts are the noise that results from motion of the electrode in relation to the subject's skin [1]. When the subject walks or runs, the waveform of the motion artifact completely overlaps the ECG signal waveform. Therefore, motion artifacts are the most difficult type of noise to eliminate. In recent years, adaptive filtering has become one of the effective and popular methods for processing and analyzing ECG signals. The adaptive filter allows the detection of time-varying potentials. In addition, they modify their behavior based on input signals. Therefore, they can detect shape changes in the whole and obtain a better signal estimate.

Several studies have been made on removing noise from ECG signals by different filtering methods based on LMS algorithm. Tong et al. [1] proposed an adaptive filter for electrode motion artifacts caused by relative motion between the subject's skin and electrode. The authors designed two motion sensors to measure electrode motion, and these two sensors are a two-axis anisotropic magneto resistive sensor and three-axis accelerometers respectively. The optimal weight of the adaptive filter is adjusted by the LMS algorithm. Mary Anne D. Raya et al. [2] used the ADXL105, a single-axis accelerometer manufactured by Analog Devices, to attach it to the waist span of the patient's back. Then, the collected acceleration signal generated by the patient walking or running is input to the adaptive filter as a noise reference signal. Finally, the parameters of the filter are adjusted by LMS and Recursive Least Squares (RLS) to minimize the error signal and then suppress motion interference. Liu et al. [3] proposed that skin strain is measured non-invasively using a light emitting diode and an optical sensor incorporated in an ECG electrode, and the results demonstrate that this device and method can effectively remove motion artifacts. Yoon et al. [4] proposed that motion artifacts reduction use a 3-axis accelerometer in e-textile ECG measurement system. And the author switched to the three-axis accelerometer module ACH-04-08-05 produced by MSI and installed in the ECG recording device to collect acceleration signals in three orthogonal directions when the human body is active. The acceleration signal is used as the reference input signal of the adaptive filter. The literature adjusts the filter parameters by normalized LMS (NLMS) to improve the correlation between the acceleration signal and the noise signal. K. Pandey [5] investigated an adaptive filtering for removal of baseline wander from the ECG signals obtained from a Holter ECG monitor. It uses a reference signal, obtained by adding absolute power spectrum of 3-channel accelerometer output connected to an ECG electrode. Thakor et al. [6] studied an adaptive filter structure for noise reduction and the reference input signal is an impulse train coinciding with the QRS complex. Shafa-at Ali Sheikh et al. [7] investigates removal of BLW from ECG signals using State Space Recursive Least Squares (SSRLS) adaptive filter and tracking of baseline wander by SSRLS has been done by using sinusoidal model in state space.

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However, the steady-state convergence analysis of the LMS algorithm with deterministic reference input indicates that the steady-state weight vector is biased, so the adaptive estimate is not close to the Wiener solution. To deal with this shortcoming, we consider using another method, the Block Least Mean Square (BLMS) algorithm, in which the coefficient vector is updated based on the block gradient estimate each time it occurs [8], [9]. The BLMS algorithm has similar characteristics to the traditional LMS algorithm, and the basic difference is that the estimation methods of the gradient vectors in the respective implementations are different. Comparing the gradient estimation of the BLMS algorithm with the conventional LMS algorithm, the BLMS algorithm uses a more accurate gradient vector estimation. Due to the time averaging, it has the characteristic that the estimation accuracy increases as the block length increases. To the best of our knowledge, BLMS algorithm has not been considered in the context of the use of accelerometer measurement data in ECG signals as a reference signal on reducing motion artifacts.

In this paper, we use a three-axis accelerometer to measure the acceleration signal of the vibration or the motion of the torso as the reference input of the adaptive filter and adjust the optimal weight of the adaptive filter by the BLMS algorithm to remove motion artifacts. Finally, we applied the algorithm to ECG signals from the subjects and compared its performance with the traditional LMS algorithm. The results show that the BLMS algorithm performs better than the LMS algorithm on eliminating ECG motion artifacts under the premise of using the three-axis accelerometer as the reference input signal. In addition, we found the adaptive filter based on BLMS algorithm can make the R wave of the filtered ECG clearly appeared.

2. MATERIALS AND METHODS

2.1 Adaptive Filter System

The adaptive filter is a filter which can use an adaptive algorithm to change the parameters and structure of the filter based on changes in the environment. In general, the structure of the adaptive filter is not changed. The coefficients of the adaptive filter are time-varying coefficients updated by the adaptive algorithm. That is, its coefficients are automatically and continuously adapted to a given signal to obtain the desired response. The most important feature of an adaptive filter is its ability to work effectively in an unknown environment and to track the time-varying characteristics of the input signal. Adaptive filter systems can reduce motion artifacts effectively and extract the useful signal in the noise. The core idea of the adaptive filter is to use an adaptive algorithm and a transversal filter to adjust the filter parameters iteratively and perform dynamic filtering on the random signal [10].

Figure 1 shows the structure of an adaptive filter system. The adaptive filter system has two input signals, one is the original input *r*, where includes the effective ECG signal s and noise signal n_0. And the other is the reference input *x* that is uncorrelated with *s*. The reference signal *x* is input to the filter and then subjected to the weight change to obtain the output signal *y*. Therefore, the adaptive filter system error signal *e* can be formulated by the effective ECG signal *s*, noise signal n_0 and the filter output signal *y* as follows:

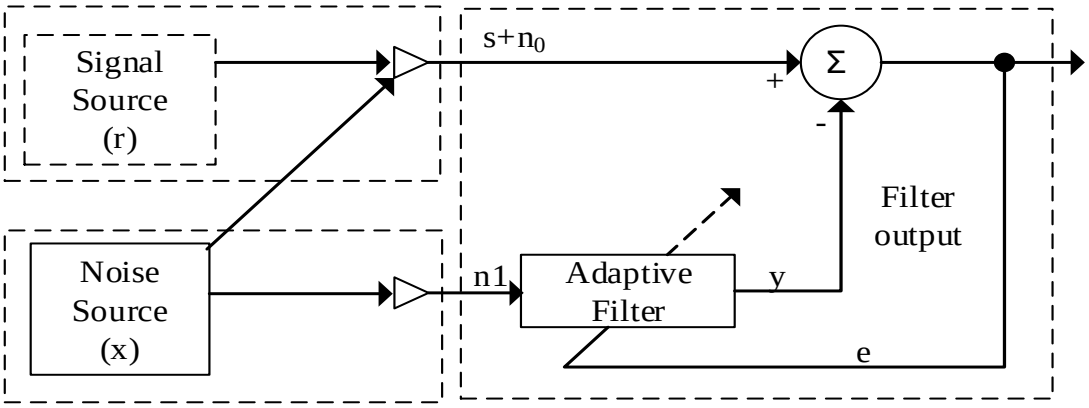


Figure 1. The conceptual structure of an adaptive filter system

$$e = s + n_0 - y \quad (1)$$

The power or energy of this signal is computed by squaring Eq(1),

$$e^2 = (s + n_0 - y)^2$$
$$= (s + n_0)^2 - 2y(s + n_0) + y^2$$
$$= (n_0 - y)^2 + s^2 + 2sn_0 - 2ys \quad (2)$$

Since the signal *s* and noise n_0 are uncorrelated, the mean-squared error (MSE) is

$$E[e^2] = E[(n_0 - y)^2] + E[s^2] \quad (3)$$

The adaptive filter extracts the signal or eliminates the noise by iteratively minimizing the MSE between the original input and the reference input. The noise in the original signal needs to be attenuated to the maximum extent, so that $E[e^2]$ reaches a minimum while ensuring that maximum the energy of the effective ECG signal is retained.

2.2 Block Least Mean Square (BLMS) Algorithm

The block diagram of the filter structure based on the BLMS adaptive algorithm is shown in Figure 2. The electrocardiographic signal *r* with motion artifact noise is used as the original input signal, where *r* includes the effective ECG signal *s* and the noise signal n_0. We use the 3-axis accelerometer MPU6050 to collect the human body acceleration signal and use it as a reference signal *x*. Filter coefficient is *w*, and adaptive filter output signal is $y(n) = xw^T$. Thus, according to the steps of the BLMS adaptive filtering algorithm, we can perform ECG signal motion artifacts filtering processing through MATLAB®R16.

Unlike filter coefficients of the traditional LMS algorithm are recalculated and updated for each sampled value. The block processing algorithm updates the weight coefficients in each block area. The implementation steps of the dynamic ECG signal motion artifacts filtering based on BLMS adaptive algorithm are as follows:

- Parameters: L is the length of block, μ is step size.
- Initialization: the weight vector w is initialized, $w(0) = 0$;
- Computation: since the error signal in the BLMS algorithm varies with the sampling rate, it can be concluded that for each data block, we have different error signal values for the adaptive process. The tap weights of the filters are updated after each block of data samples is collected, so that the adaptive filters are processed on a block basis rather than sample by sample as in the conventional LMS algorithm. the tap weight vector is updated from block to block as,

$$w(k+1) = w(k) + \mu \sum_{i=0}^{L-1} x(kL+i)e(kL+i) \quad (4)$$

A notable feature of the block BLMS algorithm is the combination of the following gradient vector estimates in its design.

$$\nabla(k) = -\frac{1}{L}\sum_{i=0}^{L-1} x(kL+i)e(kL+i) \quad (5)$$

In BLMS algorithm, the gradient estimate can vary with the length of the block, which make it more accurate than LMS algorithm.

The BLMS algorithm can be written as the following equation according to equation(5),

$$w(k+1) = w(k) - \mu_B \nabla(k) \quad (6)$$

$$\mu_B = L\,\mu \quad (7)$$

where, introducing factor 1/L is to make ∇(k) become an unbiased time average. μ_B can be regarded as an effective step size parameter of the BLMS algorithm.

And $w(k) = [w_0(k), w_1(k),\ldots, w_{L-1}(k)]^t$ is the tap weight vector corresponding to the kth block, $x(kL+i) = [x(kL+i), x(kL+i-1),\ldots, x(kL+i-L+1)]^t$ and $e(kL+i)$ is the error signal at $n = kL+i$.

The current signal error value is calculated according to the input,

$$e(kL+i) = r(kL+i) - y(kL+i) \quad (8)$$

where $r(kL+i)$ is the original ECG signal corrupted by motion artifacts and $y(kL+i)$ is the filter output signal at $n = kL+i$, given as,

$$y(kL+i) = w^t(k)x(kL+i) \quad (9)$$

- Repeat the above calculation steps to complete the set number of iterations. In paper, the number of iterations is the length of the original input signal.

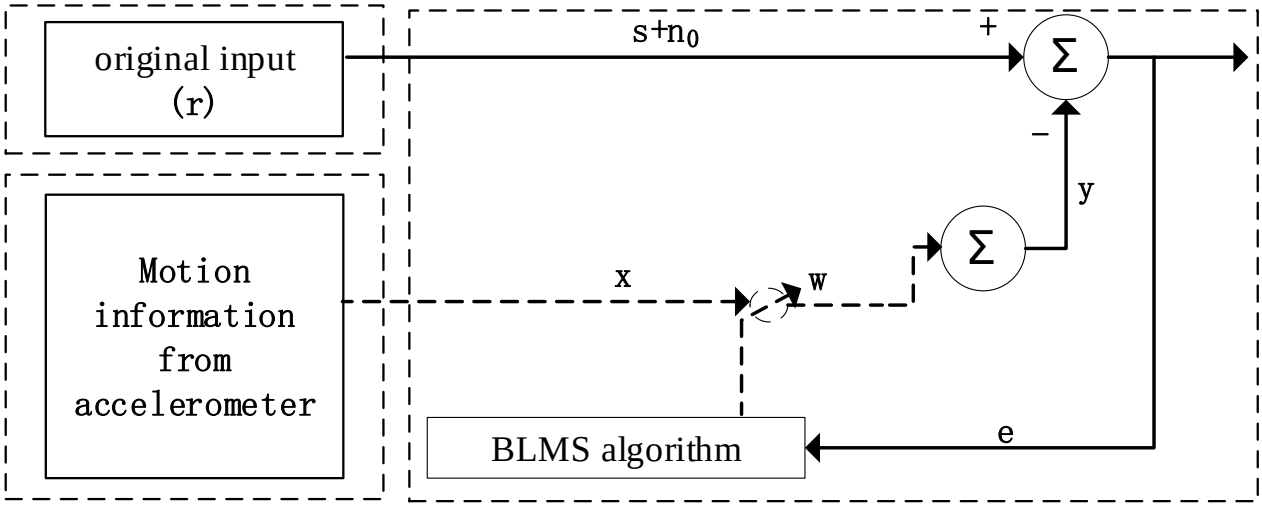


Figure 2. The block diagram of the filter structure based on BLMS adaptive algorithm

3. RESULTS

The experiment collected the ECG signals of the subjects under the two actions of sitting and bending. In the experiment, the BIOPAC® MP150 collects ECG signals through the ECG amplifier module (ECG100C), where the sampling rate of the ECG100C is set to 2KHZ. In addition, the subjects wear the fabric electrode and the center of the chest overlaps the wear acceleration module; we use the fabric electrode to collect the ECG signal and use the ECG signal as the original input signal. some researchers have proposed several methods of collecting reference signals on adaptive filtering, such as measurement of skin-electrode impedance [11], skin stretching measured with sensors [12] and accelerometers. In this experiment, we obtain the motion acceleration signal through the 3-axis piezoresistive silicon acceleration sensor module (MPU6050) and use it as reference input signals. The sampling rate of the accelerometers is set to 100HZ. For system inputs with Intel® Core™2 CPU 8550U@1.80 GHz and 8GB RAM. In this experiment, we collected 3000 samples of ECG signals. And in the simulation, μ is chosen as 0.01, and the block length is 10. The ECG signal is filtered by the human motion information collected by the 3-axis acceleration. And we utilize the adaptive filter algorithm to reduce motion artifacts. The tap weight is adaptively adjusted by the BLMS algorithm and compares with the processing results of the LMS algorithm. The performance of LMS algorithm and BLMS algorithm under static and motion states are compared respectively. The comparison results are shown in Figure 3 and Figure 4 respectively in the two states. Figure 3 shows the performance of the adaptive filter under the static state. Figure 4 shows the performance of the adaptive filter under the movement.

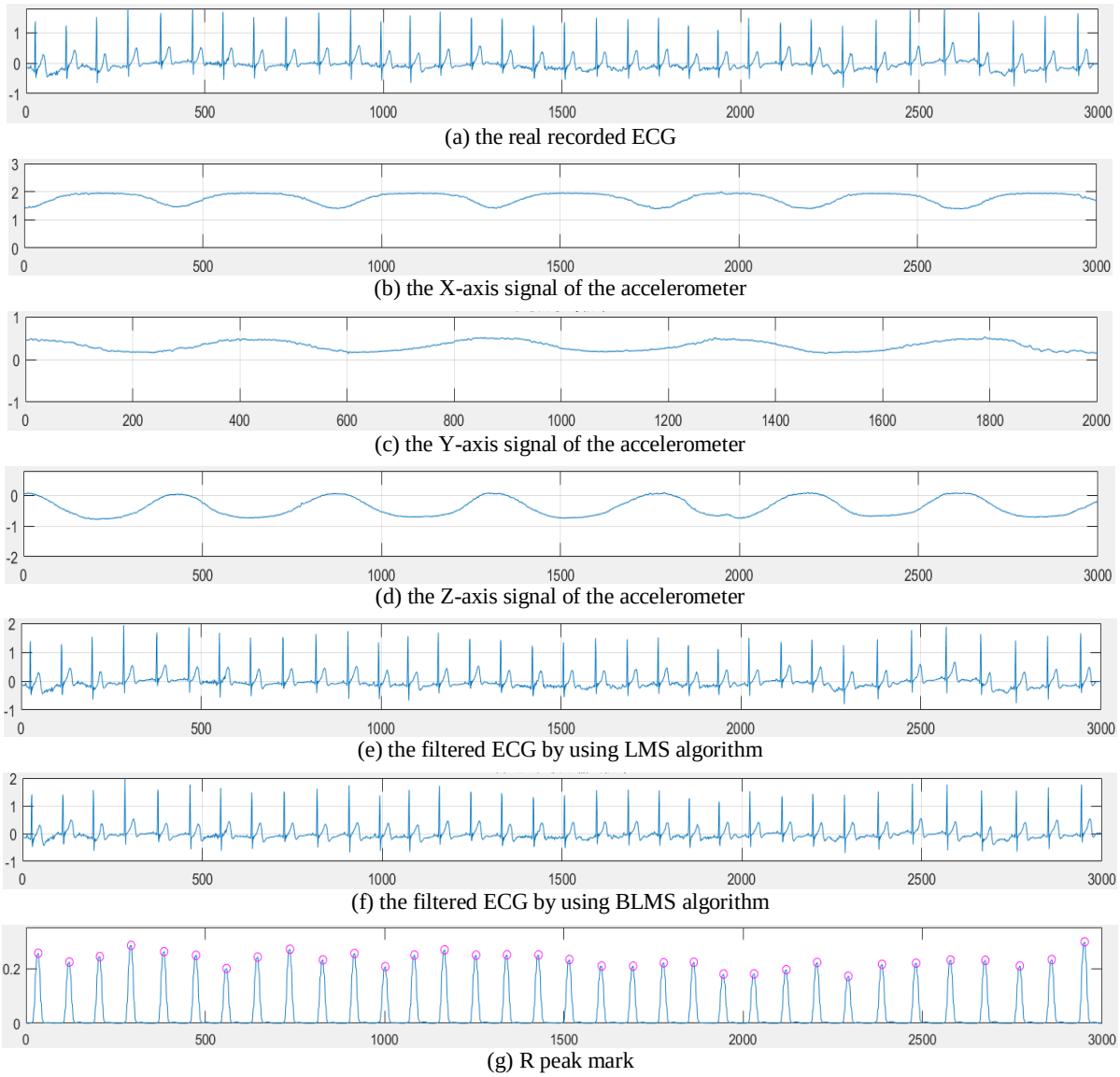


Figure 3. the performance of the adaptive filter under static state, (a) the real recorded ECG, (b) the X-axis signal of the accelerometer, (c) the Y-axis signal of the accelerometer, (d) the Z-axis signal of the accelerometer, (e) the filtered ECG by using LMS algorithm, (f) the filtered ECG by using BLMS algorithm and (g) R peak mark

It can be seen from Figures 3(b)-(d) that the acceleration signals collected by the X-axis, Y-axis and Z-axis of the three-axis accelerometer are relatively stable, and there is no large jitter and sudden change. The vibration amplitudes of the three axes are all less than 1, indicating that the human body has a small amplitude of motion in this state, and the ECG acquisition is not affected by the human motion, only a little baseline drift caused by respiration. We apply the two adaptive algorithms of LMS and BLMS respectively to the suppression of motion artifacts of ECG signals. The various bands and intervals of the center electrical signal in Figure 3(f) are clearer than Figure 3(e). Figure 3(f) is better than the removal of baseline drift in Figure 3(e). Figure 3(g) shows the R peak with a red dot.

It can be seen from Figures 4(b)-(d) that the triaxial accelerometer has obvious jitter, and the fluctuation of motion interference exceeds the R peak, which makes it difficult to detect and locate the R peak. However, there is a correlation between the motion artifact interference of the ECG signal and the human motion. Therefore, this paper adopts an adaptive filter based on BLMS algorithm to filter ECG signals and reduce motion artifacts. Figure 4(g) shows the results of R peak detection in the motion state after filtering. The R peak marks in the Figure 4(g) are clear, and each R peak can be accurately detected.

In the LMS algorithm, we determine the step size parameter as 0.01 by calculation and experimental comparison, and the number of iterations is the length of the input signal. As can be seen from Figure 4(e) and Figure 4 (f), BLMS algorithm has a better filtering effect than LMS algorithm. We apply the two adaptive algorithms of LMS and BLMS respectively to the suppression of motion artifacts of ECG signals. As can be seen from the waveform of ECG in Figure 4(a), the motion artifact interference has a great influence on the ECG waveform, which makes the ECG waveform severely distorted. The interference amplitude exceeds the R peak, and effective R peak detection cannot be performed.

Figure 4(e) shows the effect of filtering processing using the LMS adaptive algorithm. It can be seen from the figure that the motion artifact interference is still serious. Figure 4(f) is the effect diagram of motion artifact suppression using BLMS adaptive algorithm. Compared with filtering, the optimized BLMS adaptive algorithm has better filtering effect.

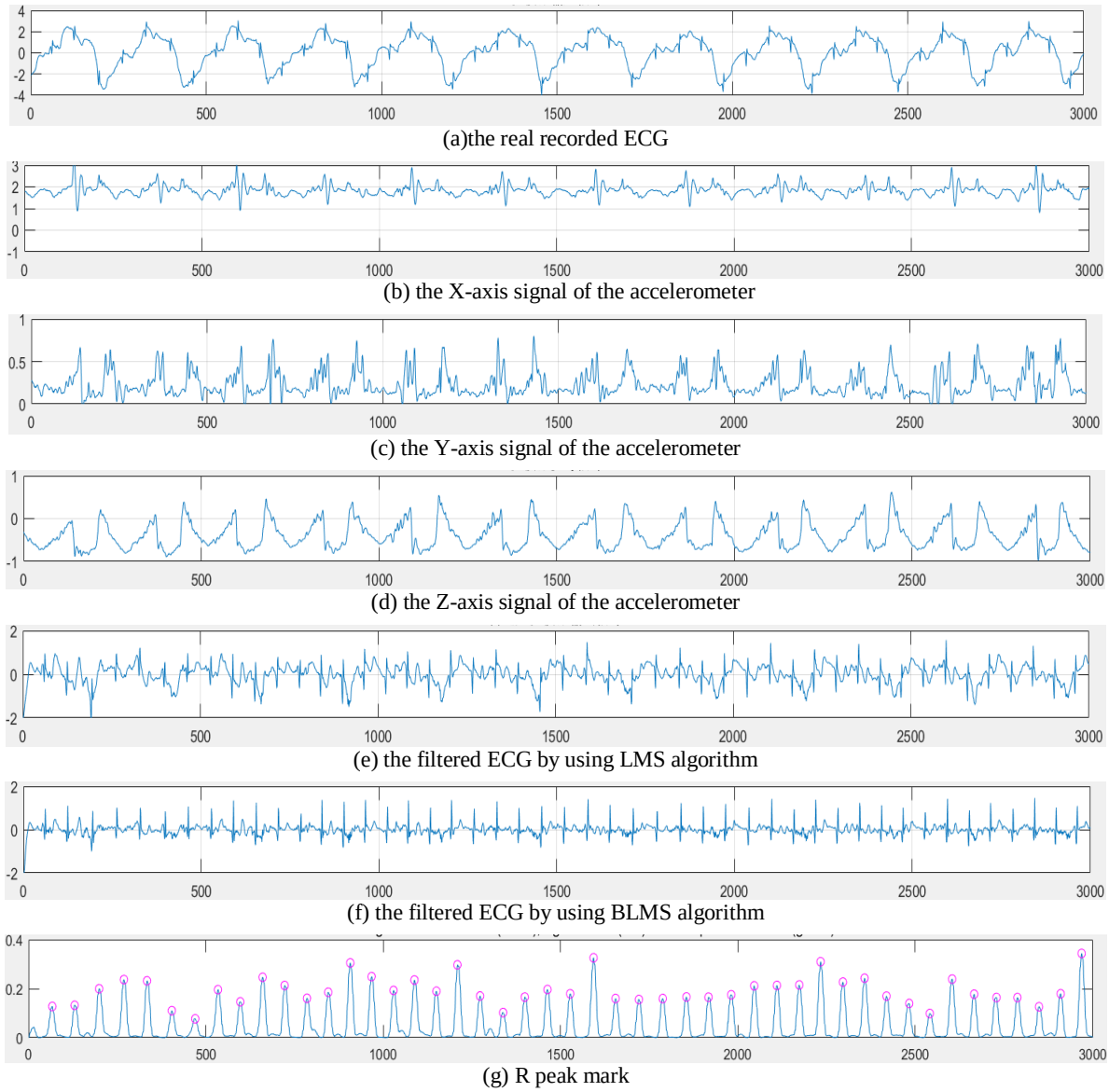


Figure 4. the performance of the adaptive filter under the movement, (a) the real recorded ECG, (b) the X-axis signal of the accelerometer, (c) the Y-axis signal of the accelerometer, (d) the Z-axis signal of the accelerometer, (e) the filtered ECG by using LMS algorithm, (f) the filtered ECG by using BLMS algorithm and (g) R peak mark

4. CONCLUSION

In this paper, we describe the estimation and reduction of motion artifacts by using the BLMS algorithm, which minimizes the mean square error based on the steepest descent strategy and block gradient estimation and uses motion acceleration from a 3-axis accelerometer as the reference input signal. In BLMS algorithm, the gradient estimate can vary with the length of the block, which make it more accurate than LMS algorithm. Using an axis reference input, especially the y-axis, is enough to minimize noise. This is because the main component of motion acceleration during human motion is usually found in the vertical direction. Finally, we found that the performance of the BLMS algorithm is better than the LMS algorithm.

5. ACKNOWLEDGMENTS

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